PROJECT DESIGN - SYSTEM ARCHITECTURE

Courier

Courier Management - 3 Tier Architecture

TIER 1: PRESENTATION LAYER (FRONTEND)

- Tech: HTML5, CSS3, Bootstrap 5, React.js / JavaScript
- Pages: Home Page, Booking Form, Tracking Page, Login, Register
- Admin: Dashboard, Agent Assignment, Reports, Invoice View

[] REST API - JSON []

TIER 2: APPLICATION LAYER (BACKEND)

- Framework: Python Flask / Java Spring Boot / PHP
- **Modules:**
  1. User Module - Login, Register, OTP Verification
  2. Booking Module - Cost Calculation, Tracking ID
  3. Tracking Module - Status Update
  4. Delivery Module - Agent + POD (Photo/OTP)

[] SQL Queries / ORM []

TIER 3: DATA LAYER (DATABASE - MySQL)

- **Tables (5 Tables):**

USERS (user_id, name, email, role)

TRACKING_HISTORY (history_id, status, location)

SHIPMENTS (shipment_id, sender_id, tracking_id, status, cost)

AGENTS, INVOICES, POD Tables | 3NF Normalized

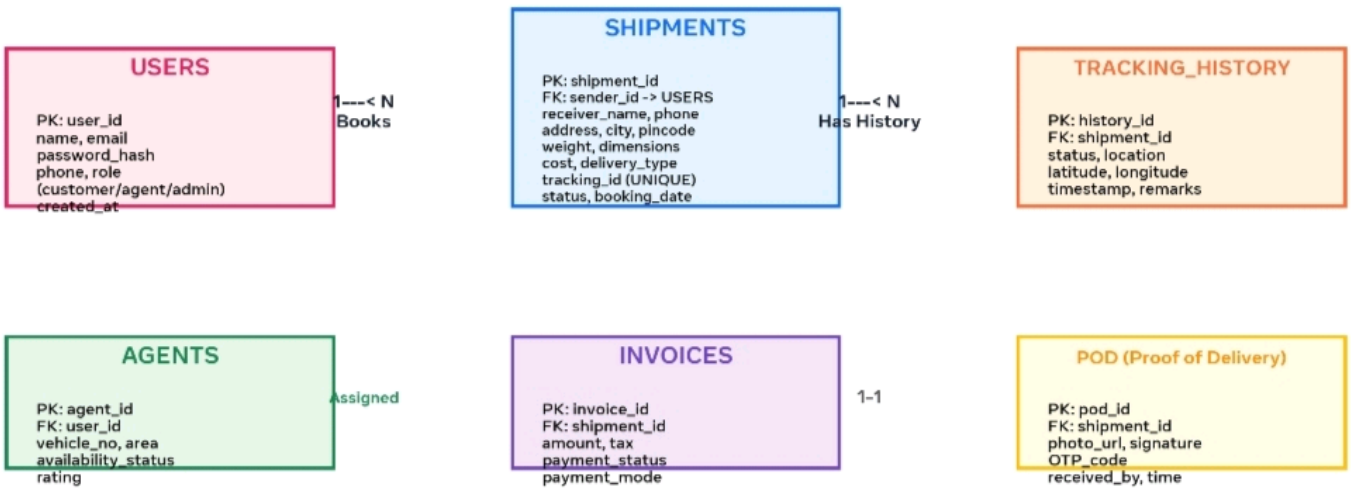
API List:

```
POST /api/register   |   POST /api/login   |   POST /api/book
GET /api/track/{id}   |   PUT /api/update-status   |   GET /api/invoice/{id}
```

Tracking ID Format: CMS + 6 Digit Number (Ex: CMS123456)

Status Flow: Booked -> Picked -> In-Transit -> Out for Delivery -> Delivered

PHASE 3: PROJECT DESIGN - ER DIAGRAM & DATABASE DESIGN



Database Design: MySQL, 3NF Normalized, Primary Keys, Foreign Keys, Indexes on tracking_id, user_id

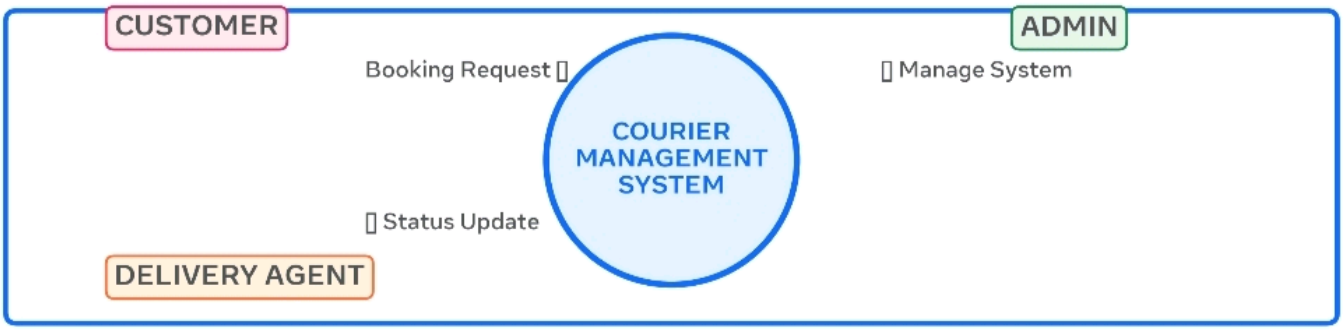
Normalization:
- 1NF: All atomic values (no repeating groups)
- 2NF: Full dependency on PK
- 3NF: No transitive dependency

ER Cardinality:
- Users (1) -> Shipments (N)
- Shipments (1) -> Tracking_History (N)
- Shipments (1) -> Invoice (1)
- Shipments (1) -> POD (1)
- Users (1) -> Agents (1)

Indexes: tracking_id UNIQUE, sender_id, shipment_id in tracking_history for fast tracking queries

PROJECT DESIGN - DFD, UML & WIREFRAMES

DFD Level 0 - Context Diagram



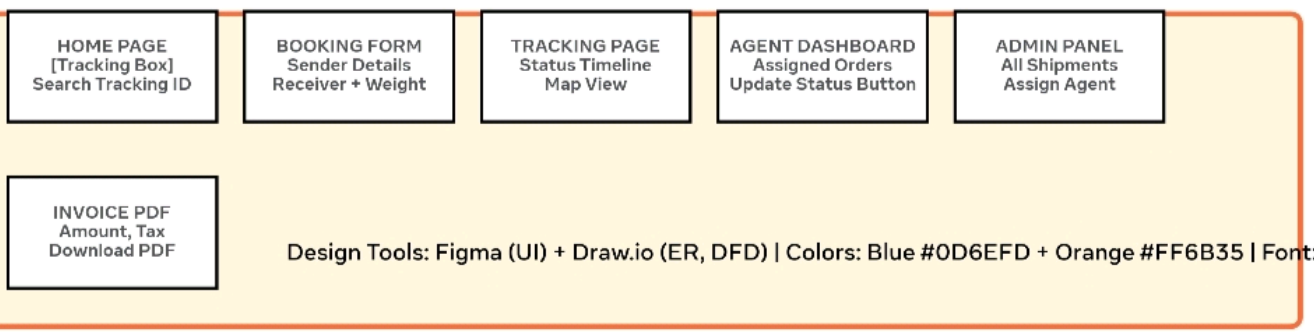
DFD Level 1 - Detailed Process

- **1.0** BOOK PARCEL - Get sender, receiver, weight, address -> Save to SHIPMENTS table
- **2.0** CALCULATE COST - Weight * Distance + Delivery Type -> Generate Tracking ID CMS123456
- **3.0** ASSIGN AGENT - Admin assigns nearest available agent -> Send SMS/Email to agent
- **4.0** TRACKING UPDATE - Agent updates: Picked -> In-Transit -> Out for Delivery -> Delivered

UML Class Diagram

```
class User {
   user_id: int, name: string, email: string, role: string
   methods: register(), login(), logout()
}
class Shipment { shipment_id, tracking_id: CMS123456, weight, cost, status }
   methods: bookParcel(), calculateCost(), generateTrackingId(), trackShipment()
```

Wireframes - UI Design (Figma)



Design Tools: Figma (UI) + Draw.io (ER, DFD) | Colors: Blue #0D6EFD + Orange #FF6B35 | Font: Poppins